# Global Equity Market Dependence at Two Horizons

## Measurement gaps, identification boundaries and portfolio implications

August 2026 · MSc Finance with Data Science, UCL · 14 markets, 2005–2025 · LSEG Workspace data

**FINDING** In this panel, daily and weekly returns measure systematically different dependence, and the difference is concentrated in pairs of markets that close at different times. The design identifies the gap, not its causal decomposition. Whether the gap changes realised portfolio risk depends on the portfolio's constraints.

### DESIGN

- Fourteen local-currency equity indices (Americas, EMEA, Asia-Pacific): 3,579 common trading days, 1,056 weeks, 91 pairs. Weekly returns are built from each market's last close in each ISO week.
- Three dependence measures: correlation stratified by time-zone relationship, generalized forecast-error-variance connectedness, and DCC conditional correlation.
- Five questions registered before estimation in version-controlled configuration: Q1 size of the gap, Q2 one-day realignment, Q3 portfolio consequence, Q4 centrality ranking, Q5 common component.

### Q1 · THE MEASUREMENT GAP

| | Daily | Weekly | W – D | 95% CI |
|---|---|---|---|---|
| Cross-time-zone correlation (63 pairs) | 0.286 | 0.456 | +0.170 | +0.142, +0.195 |
| Same-time-zone correlation (28 pairs) | 0.430 | 0.473 | +0.043 | |
| Total connectedness (%) | 60.40 | 72.56 | +12.16 | +9.72, +13.85 |
| Within-to-cross regional ratio | 1.87 | 1.12 | −0.75 | |

The gap is positive for all 63 cross-zone pairs against 22 of 28 same-zone pairs; the cross-minus-same contrast is +0.110 to +0.143, and none of 500 block-bootstrap draws reverses the ordering. United States and Japan: 0.149 daily, 0.646 weekly. The calendar-matched horizon comparison gives the same connectedness levels.

### IDENTIFICATION

| Pairwise gap on closing-time distance (91 dyads) | Baseline | Full controls |
|---|---|---|
| Closing-time distance (hours) | +0.0181 (0.0034) | +0.0166 (0.0046) |
| QAP permutation p (5,000 relabelings) | 0.0002 | 0.0010 |
| Log great-circle distance | | +0.0057 (0.0162) |
| $R^2$ | 0.528 | 0.541 |

Two-way dyadic cluster-robust standard errors. Closing time survives every control (t = 3.6); distance does not (t = 0.35). Q2: shifting Asian returns one day raises Americas–Asia correlation by only 0.0047 but lowers EU/MEA–Asia by 0.182, so the global remedy fails.

### Q3 · PORTFOLIO CONSEQUENCE

| Minimum-variance, annualised volatility (pp) | Daily engine | Weekly engine | D – W |
|---|---|---|---|
| Forecast volatility | 7.99 | 8.57 | −0.58 |
| Realized volatility | 10.21 | 10.37 | −0.16 |
| Understatement (realized – forecast) | 2.22 | 1.80 | +0.42 |

| At a matched 24-month estimation span | D – W | 95% CI |
|---|---|---|
| No position cap (3.6 effective holdings) | −0.02 | −0.61, +0.52 |
| Change when a 10% cap is imposed (10.7 holdings) | +0.59 | +0.15, +1.13 |

Rebalanced every four weeks, evaluated weekly. The raw 0.42 confounds sampling frequency with a 12- versus 24-month window. The capped level itself (+0.57) has an interval that includes zero, so the evidence supports a constraint-dependent change, not a precisely estimated cost. Diebold–Mariano rejects equal accuracy in one of 36 tests, which does not survive a multiplicity correction.

### Q4 AND Q5 · NETWORK READING AND THE COMMON COMPONENT

- The United Kingdom and Germany are the largest net transmitters at both frequencies (rank correlation 0.855), but middle rankings move with the horizon.
- Removing the first principal component reduces median connectedness from 60.06 to 32.91 daily and from 72.33 to 22.62 weekly. The component takes 40.1% of standardized daily variance and 53.1% weekly; it is statistical, with no pricing interpretation, and what remains is still connected (34.13% daily, 25.31% weekly).

### PRACTICAL RULES

- Compare covariance engines over equal calendar spans.
- Evaluate risk at the horizon the mandate is judged on.
- Expect constraints to make covariance structure economically consequential.
- **Implication, not tested here:** private holdings marked quarterly with smoothed valuations face a related stale-pricing problem, so diversification read from NAV marks deserves caution and holdings should be stressed together.

### LIMITS

Close-to-close data cannot separate non-synchronous closing from genuine horizon dependence; common-window intraday returns would be needed. The magnitude is panel-specific, weekly tests have a third of the sample and less power, and the constrained cost levels remain imprecise. Daily VaR: unconditional coverage is not rejected, independence is rejected; weekly, neither test rejects.

**Data and code.** LSEG Workspace local-currency price indices, 2005-01-03 to 2025-12-31. Python pipeline with pre-registered configuration files and audit scripts that tie the reported figures to stored output.

**Status.** Academic work submitted for the MSc Finance with Data Science at UCL. Not investment advice. Full dissertation, unabridged manuscript and code available on request.

Liang-Yu Lai · liangyu.chloestine.lai@gmail.com · linkedin.com/in/liang-yu-lai · xchl88.github.io